

Mathematics Reference: Percentages

Educational Series

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1 Introduction

A percentage is a number or ratio expressed as a fraction of 100. The word comes from the Latin *per centum*, meaning “by the hundred.” It is denoted using the percent sign, “%”.

2 Fundamental Formulas

2.1 Basic Definition

To find the percentage of a value:

$$\text{Percentage} = \left(\frac{\text{Value}}{\text{Total Value}} \right) \times 100$$

2.2 Percentage Change

When a value increases or decreases, the change is calculated as:

$$\text{Percentage Change} = \left(\frac{\text{Final Value} - \text{Initial Value}}{\text{Initial Value}} \right) \times 100$$

- A positive result indicates a **Percentage Increase**.
- A negative result indicates a **Percentage Decrease**.

3 Conversion Rules

3.1 Fraction to Percentage

Multiply the fraction by 100 and add the % sign.

$$\frac{4}{5} \rightarrow \left(\frac{4}{5} \times 100 \right) \% = 80\%$$

3.2 Percentage to Fraction

Remove the % sign and divide by 100.

$$25\% \rightarrow \frac{25}{100} = \frac{1}{4}$$

4 Successive Percentage Changes

If a value is changed by $x\%$ and then by $y\%$, the effective net percentage change is given by the formula:

$$\text{Net Change} = \left(x + y + \frac{xy}{100} \right) \%$$

Note: Use a negative sign for a percentage decrease.

5 Common Percentage-Fraction Equivalents

Percentage	Fraction	Percentage	Fraction
10%	1/10	50%	1/2
12.5%	1/8	60%	3/5
20%	1/5	66.66%	2/3
25%	1/4	75%	3/4
33.33%	1/3	80%	4/5

6 Product Stability Formula

If the price of a commodity increases by $r\%$, then the reduction in consumption so as not to increase the expenditure is:

$$\text{Reduction \%} = \left(\frac{r}{100 + r} \times 100 \right) \%$$

Conversely, if the price decreases by $r\%$:

$$\text{Increase in Consumption \%} = \left(\frac{r}{100 - r} \times 100 \right) \%$$

7 Practice Example

Example: The price of sugar rises by 25%. By how much percent must a housewife reduce her consumption so that the expenditure remains the same?

Solution: Using the formula:

$$\text{Reduction} = \left(\frac{25}{100 + 25} \times 100 \right) \% = \left(\frac{25}{125} \times 100 \right) \% = 20\%$$